

Exploring GFlowNets for Analog EDA

A first mathematical and computational study of diverse Pareto search

Technical project report

Dr Jesus Ocariz

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Motivation

My background is in mathematics and scientific machine learning rather than circuit design. I completed a PhD in Mathematics on nonlinear partial differential equations and geometric analysis, followed by master's-level work in applied mathematics, computer engineering and data science. My data-science thesis and subsequent R&D work at Sony Europe focused on multi-objective Generative Flow Networks (GFlowNets). This project is my attempt to learn the fundamentals of analog electronic design automation (EDA) through a concrete, reproducible experiment and to ask where that GFlowNet expertise could be useful.

GFlowNets are attractive when the goal is not one optimum but a distribution of diverse, high-reward structured objects. That idea has been explored in molecular and drug discovery, where many chemically distinct candidates may be valuable, and in Neural Architecture Search (NAS), where multiple architectures can occupy different points on an accuracy-complexity frontier [1–3]. Analog design appears to have a related mathematical structure: circuit evaluation is expensive, variables interact nonlinearly, feasibility matters, and designers often need several compromises among gain, speed, power, area and robustness.

I therefore treat EDA as a new application domain, not as something this small experiment has solved. The aim is to test a narrow hypothesis with honest baselines:

Can a GFlowNet contribute useful diverse-search behaviour to a multi-objective analog design loop, either as a standalone sampler or as a learned mutation operator inside NSGA-II?

Circuit and design problem

I chose a resistively loaded common-source NMOS (N-type Metal-Oxide-Semiconductor) amplifier because it is simple enough to understand and simulate quickly, but still exhibits meaningful trade-offs. Six variables each take five discrete values, giving $5^6 = 15,625$ configurations.

Variable	Values	Main effect represented
W (μm)	2, 4, 8, 16, 32	Transconductance, capacitance, area
L (μm)	0.18, 0.27, 0.36, 0.54, 0.72	Output resistance, speed, area
Bias target (μA)	20, 40, 80, 160, 320	Gain, bandwidth, power
R_D ($\text{k}\Omega$)	2, 4, 8, 16, 32	Gain, output pole, headroom
C_L (pF)	0.2, 0.5, 1, 2, 5	Bandwidth and area proxy
V_{DD} (V)	1.2, 1.5, 1.8, 2.1, 2.5	Headroom and power

For each selected configuration, ngspice 47 performs an operating-point analysis and an AC sweep from 10 Hz to 10 GHz. The simulation returns low-frequency voltage gain, -3 dB bandwidth, DC power and an operating-validity flag. Invalid designs cannot enter the Pareto archive. Results are cached by the six integer design indices, so repeated configurations never consume another simulation.

The MOSFET (Metal-Oxide-Semiconductor Field-Effect Transistor) is represented by a portable Level-1 model. This makes the search pipeline easy to reproduce without proprietary technology data, but it is deliberately modest: the numerical values should not be interpreted as predictions for fabricated silicon.

Data and objective space

The data are generated by the simulator, not taken from a pre-existing database. A scrambled Sobol sequence fixes a 1,024-configuration reference set before method comparison. It defines stable normalisation ranges and an estimated reference front. Each optimisation run then starts with 24 observations and adds six batches of 12, for a total budget of 96 ngspice evaluations.

The four objectives are written in minimisation form,

$$f(x) = (-G(x), -B(x), P(x), A_{\text{proxy}}(x)), \quad A_{\text{proxy}} = WL + 0.20R_D + 1.50C_L.$$

Trade-offs in the locked 1,024-design ngspice reference set

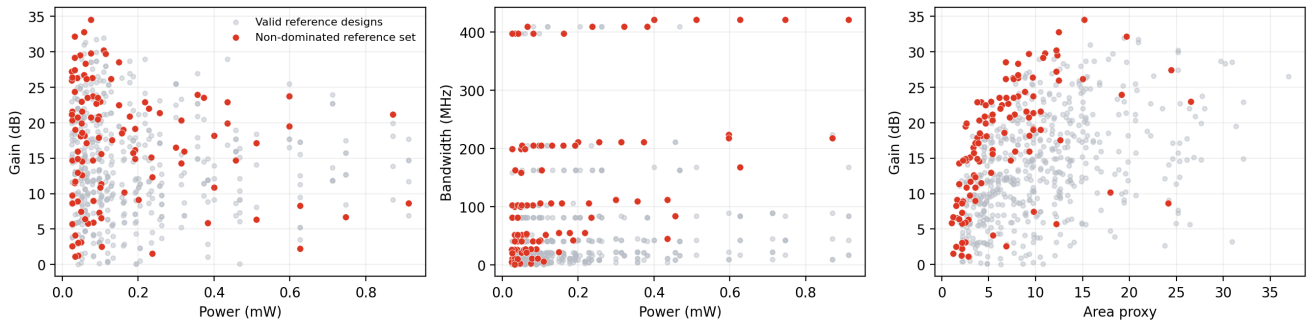


Figure 1: Projections of the locked 1,024-design reference set. Red points are non-dominated in the complete four-dimensional objective space.

Here G is gain, B is bandwidth and P is power. The final term is a transparent technology-independent proxy used to introduce an area trade-off; it is not physical post-layout area.

Methods compared

All adaptive methods use the same seven-model Extra Trees surrogate ensemble. Random Chebyshev preferences expose different parts of the front, while uncertainty and predicted operating validity contribute to acquisition.

Random and Sobol

Uniform and low-discrepancy space-filling baselines.

Preference scalarisation

Diverse greedy selection across 64 random objective preferences.

Categorical reward

Direct weighted sampling from exactly the same positive terminal reward used by the standalone GFlowNet. This tests whether performance comes from the reward rather than flow learning.

Surrogate NSGA-II

Non-dominated sorting, crowding distance, tournament selection, crossover and discrete mutation operate on surrogate predictions, following the main components of NSGA-II [4]. Only the selected batch is evaluated with ngspice.

Standalone Pareto GFlowNet

A Trajectory-Balance policy [5] sequentially assigns the six variables. It is updated for 250 steps per acquisition round and samples terminal designs in proportion to the multi-objective acquisition reward.

GFlowNet-NSGA-II feedback

This is the main proposed method. The GFlowNet persists between rounds. Designs that have been evaluated by ngspice and remain non-dominated receive an explicit reward boost. High-reward GFlowNet samples then act as a third parent—an informed mutation donor—inside NSGA-II. The selected offspring are evaluated by ngspice, successful designs update the archive, and that archive trains the GFlowNet in the next round.

The last method creates a feedback loop: NSGA-II contributes selection pressure and local evolutionary search; the GFlowNet learns a reusable distribution from successful designs; and its samples propose structured mutations rather than independent random gene changes.

Evaluation

Hypervolume measures how much of the normalised reference box is dominated by the discovered archive; higher is better. It is estimated with 100,000 fixed common Monte Carlo points. Inverted Generational Distance (IGD) measures distance from the estimated reference front to the discovered set; lower is better. Archive size counts non-dominated designs. Objective spacing is the standard deviation of nearest-neighbour distances on the normalised front, so lower values indicate more uniform coverage.

Every method uses the same five seeds, initial budget, batch sizes, reference set and ngspice cache. Reported values are mean $\pm$ sample standard deviation at 96 evaluations.

Results

Method	Hypervolume $\uparrow$	IGD $\downarrow$	Non-dominated $\uparrow$	Spacing $\downarrow$
GFlowNet-NSGA-II	0.526 $\pm$ 0.046	0.161 $\pm$ 0.008	32.6 $\pm$ 4.3	0.131 $\pm$ 0.013
Surrogate NSGA-II	0.520 $\pm$ 0.053	0.164 $\pm$ 0.016	33.0 $\pm$ 1.9	0.138 $\pm$ 0.024
Scalarisation	0.461 $\pm$ 0.032	0.175 $\pm$ 0.007	37.2 $\pm$ 5.5	0.108 $\pm$ 0.043
Categorical reward	0.410 $\pm$ 0.056	0.179 $\pm$ 0.017	39.4 $\pm$ 4.7	0.110 $\pm$ 0.035
Pareto GFlowNet	0.401 $\pm$ 0.045	0.178 $\pm$ 0.013	41.6 $\pm$ 4.3	0.097 $\pm$ 0.026
Sobol	0.370 $\pm$ 0.048	0.184 $\pm$ 0.011	28.0 $\pm$ 1.0	0.124 $\pm$ 0.025
Random	0.290 $\pm$ 0.067	0.197 $\pm$ 0.016	27.4 $\pm$ 3.0	0.139 $\pm$ 0.022

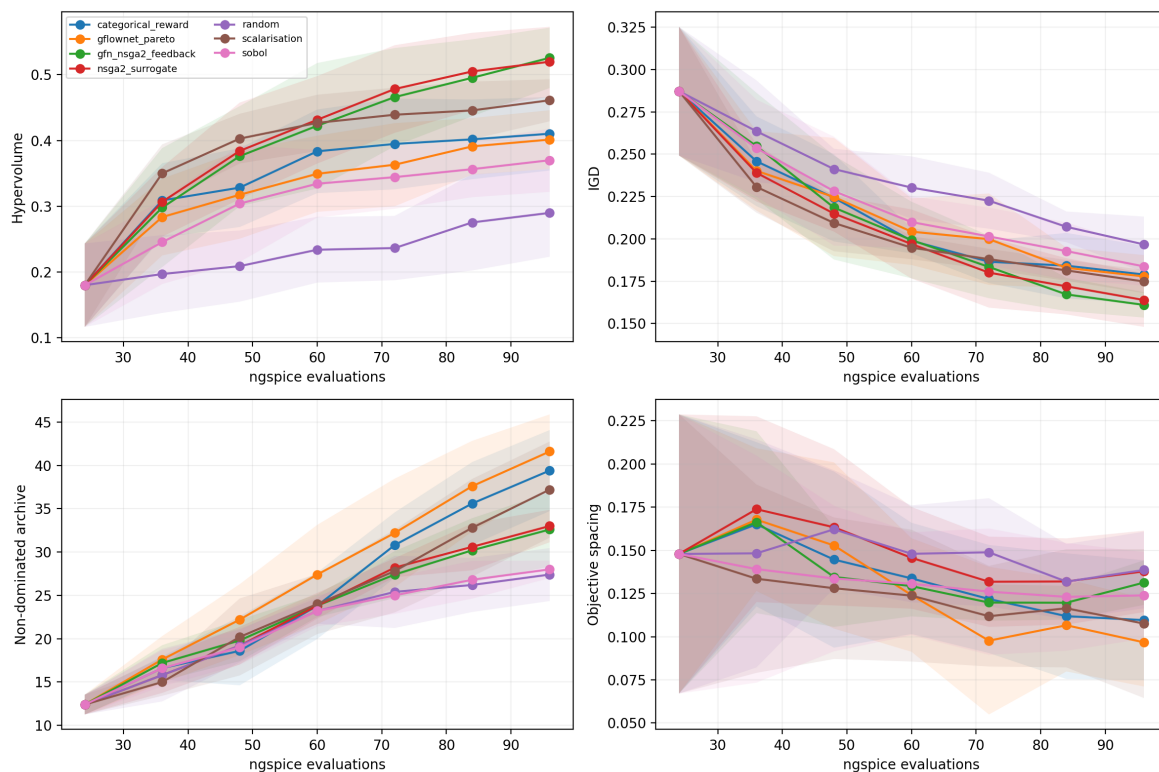


Figure 2: Pareto metrics versus ngspice-call budget. Shading represents one sample standard deviation across five seeds.

The feedback hybrid has the best mean hypervolume and IGD, improving on NSGA-II by 0.006 and 0.003 respectively. This is encouraging, but seed-level differences change sign. Paired Wilcoxon tests give $p = 0.8125$ for hypervolume and $p = 1.0$ for IGD. With five seeds, I therefore regard the hybrid as competitive with NSGA-II, not proven superior.

The standalone GFlowNet shows a different strength. It produces the largest non-dominated archive and the lowest spacing statistic, suggesting more numerous and uniformly distributed trade-offs. However, direct categorical sampling from the same reward attains slightly higher hypervolume. The present experiment therefore does not show that flow learning alone improves terminal Pareto quality.

What I learned and next steps

The experiment suggests that GFlowNets may be more useful as a component of an EDA optimiser than as a replacement. NSGA-II provides strong convergence and diversity mechanisms; the GFlowNet can learn reusable structured mutations from successful simulations. The hybrid method should be analysed with a larger study.

Selected references

- [1] Y. Bengio et al., *GFlowNet Foundations*, 2021. arXiv:2111.09266.
- [2] A. C. Nica et al., *Evaluating Generalization in GFlowNets for Molecule Design*, MLDD at ICLR 2022.
- [3] D. Soin et al., *Order-Preserving GFlowNets*, 2024; experiments include molecule generation, multi-objective Pareto approximation and neural architecture search. OpenReview.
- [4] K. Deb et al., *A Fast and Elitist Multiobjective Genetic Algorithm: NSGA-II*, IEEE Transactions on Evolutionary Computation, 2002.
- [5] N. Malkin et al., *Trajectory Balance: Improved Credit Assignment in GFlowNets*, NeurIPS 2022. OpenReview.